

Common Seizure Mimics

Everything that shakes isn't a seizure. Before you determine the patient has experienced epileptic seizures, make sure to have considered relevant seizure mimics. These nonepileptic conditions imitate epileptic seizures and recurrences may be misdiagnosed as epilepsy.

A careful history, physical examination, and video EEG evaluation are useful in diagnosing the cause of paroxysmal "seizure-like" events. Less commonly, both epileptic and nonepileptic events may coexist, necessitating inquiry into whether the event in question is consistent with the patient's typical event and if other uncharacterized events also exist. A complete evaluation should aim to capture and characterize each of the patient's event type, though frequent events are easier to characterize than those occurring seldomly.

In a majority of conditions that imitate epileptic seizures, the EEG remains unchanged during the event. However, some nonepileptic conditions are associated with abnormal EEG patterns that should not be mistaken for ictal activity. This chapter lists common seizure mimics and their associated electrographic features.

Seizure mimics are grouped as physiological (organic disease) or psychological. Furthermore, physiological mimics may be neurologic, systemic, or both (such as syncope) [1].

Seizure Mimics

Neurological Mimics

- syncope
- transient ischemic attack (TIA)
- migraines
- parasomnias
- movement disorders

Systemic Mimics

- cardiac arrhythmias
- transient toxicities and metabolic disturbances (such as hypoglycemia)

Psychological Mimics

- nonepileptic seizures and other conversion disorders
- panic attacks
- other behavioral, psychiatric, and psychological conditions

Neurological Seizure Mimics

Syncope

Syncope is characterized by a transient loss of consciousness and postural tone due to transitory global cerebral hypoperfusion. Often, there is accompanying brief multifocal tonic activity, eye deviation, vocalizations, hallucinations, and sphincteric disturbances (convulsive syncope) that can be mistaken for an epileptic seizure. Semiologically, syncopal events are brief, positional, and lack postictal confusion unlike epileptic seizures. Incontinence and lateral tongue bites are rare. Neurogenic (autonomic) and cardiogenic causes exist with vasovagal syncope being the most common cause.

The EEG during a syncope is characterized by emergence of generalized theta-delta slowing followed by a period of diffuse attenuation corresponding with transient cerebral hypoperfusion and resultant loss of consciousness. Recovery is characterized by the reemergence of generalized slowing followed by a return of the resting background. The EKG channel shows transient bradycardia or asystole [2]. Figures 18.1 to 18.4 show the sequence of EEG changes with syncope.

Rarely, epileptic seizures induce ictal or postictal cardiac arrhythmias resulting in loss of consciousness [3]. Figure 18.5 shows EEG in postictal cardiac arrest.

Transient Ischemic Attacks (TIA)

Cerebrovascular accidents such as TIAs and strokes, unlike epileptic seizures are typically associated with negative symptoms (e.g., paralysis), though occasionally positive symptoms such as limb shaking occur, especially in the setting of bilateral carotid stenosis (limb shaking TIA) which may be mistaken for an epileptic seizure. The EEG is characterized by acute onset of focal slowing corresponding to the region of cerebral ischemia [4].

Migraines

Migraines with aura can mimic epileptic seizures. Migraine and epilepsy commonly coexist further complicating the diagnosis of individual events. Migraine

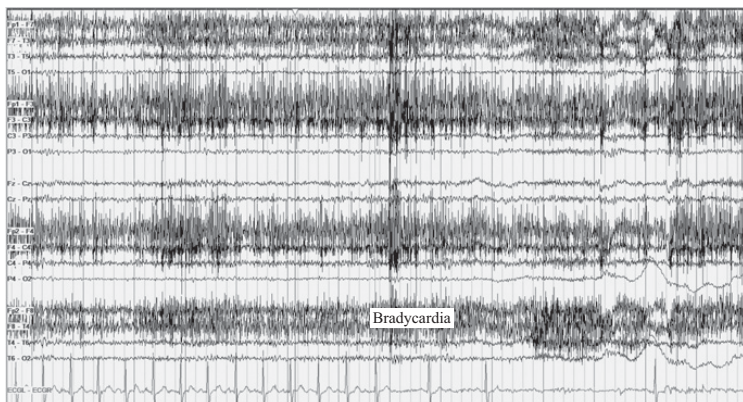


Figure 18.1 Fifty-three-year-old woman with convulsive syncope, sudden onset of bradycardia on EKG channel heralds the syncopal attack.

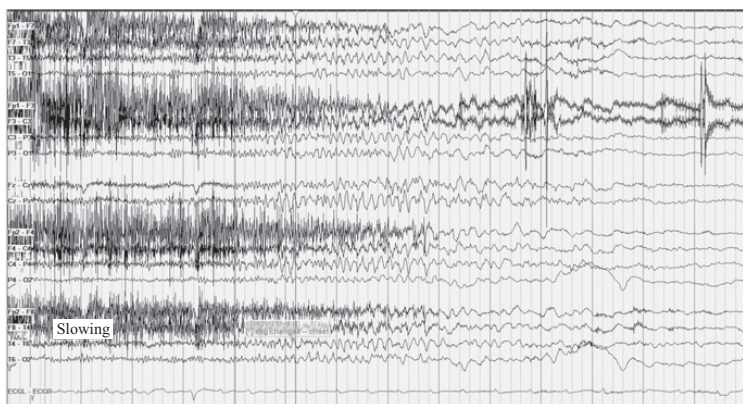


Figure 18.2 Same patient as in Figure 18.1, severe bradycardia on EKG followed by diffuse higher amplitude theta and delta slowing followed by attenuation.

may be associated with transient visual phenomena, focal motor, language, and sensory deficits and rarely in confusion, and coma. Sensory illusions and perceptual changes with hearing or vision (such as dramatic shrinking of body parts as a feature of “Alice in wonderland” syndrome) may also occur.

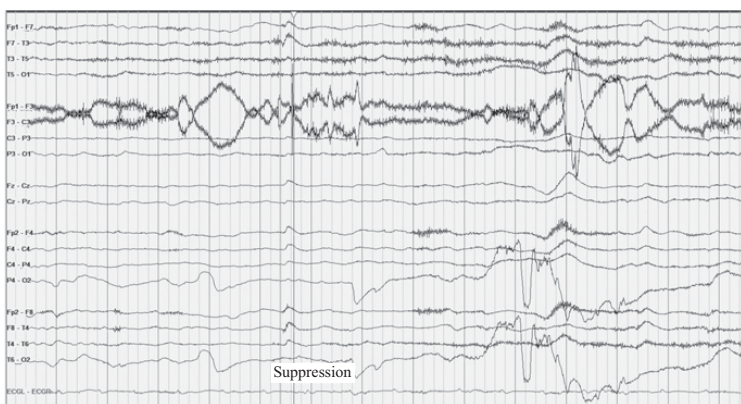


Figure 18.3 Same patient as in Figure 18.1, generalized EEG suppression (GES) with cardiac asystole.

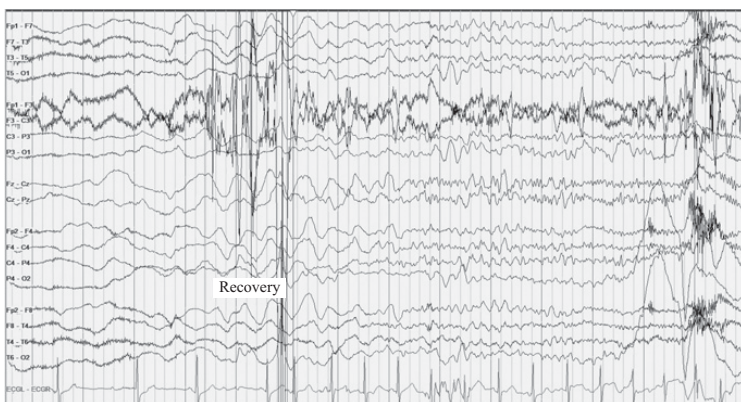


Figure 18.4 Same patient as in Figure 18.1, resolution of syncope characterized by diffuse slowing followed by rapid restoration of EEG background.

The EEG during migraines is usually normal but may show slowing (focal or diffuse) or amplitude attenuation (Figure 18.6). Though nonspecific, migraineurs have a higher incidence of photic driving [5,6].

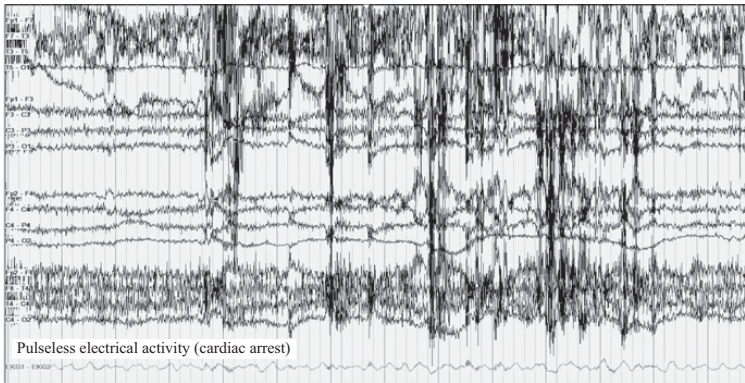


Figure 18.5 Fifty-year-old woman with postictal asystole, GES with pulseless electrical activity on EKG.

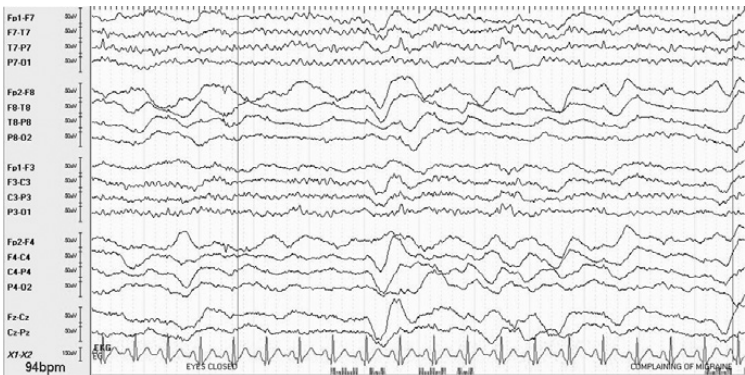


Figure 18.6 Thirty-year-old man with hemiplegic migraine, right hemisphere–predominant polymorphic delta slowing. Additionally, there is bilateral EKG artifact.

Parasomnias

Episodic sleep disorders (parasomnias) mimic nocturnal seizures. Epileptic seizures rarely occur during rapid eye movement (REM) sleep while parasomnias are specific to either REM or non-REM (slow-wave) sleep. The EEG can identify ictal activity and the sleep stage associated with the event. Sleep talking (somniloquy), sleep talking (somnambulism), confusional arousals,

and night terrors occur during slow-wave sleep while REM behavior disorder and nightmares occur during REM sleep [7].

Movement Disorders

Movement disorders including tics, tremor, paroxysmal dyskinesias, and nonepileptic myoclonus are common seizure mimics. The EEG activity during these events is normal but there is rhythmic appearing movement artifact [7].

Systemic Seizure Mimics

Cardiac Arrhythmias

Arrhythmias related cardiogenic syncope are important to recognize as they may be life threatening. In long QT syndromes ventricular tachycardias may be triggered by emotions or exercise. Syncope in sleep, sensorineural deafness, and a family history of sudden death or drowning are important clues to this disorder. EKG channel may show abnormalities in heart rhythm [8].

Transient Toxicities or Metabolic Disturbances

Toxic and metabolic disturbances that lead to delirium, myoclonus, or transient deficits may mimic or precipitate seizure activity. The EEG usually shows nonspecific encephalopathy (Chapter 23 describes the EEG in encephalopathy). Hypoglycemia in particular, accentuates hyperventilation build up [8,9].

Psychological Seizure Mimics

Nonepileptic Seizures

Nonepileptic seizures (previously termed psychogenic seizures, nonepileptic attacks, or pseudoseizures) are behavioral events that resemble epileptic seizures without an electrographic correlate or clinical evidence of epilepsy. There may be a history of psychiatric comorbidities, emotional, or physical trauma. They commonly manifest with seizure-like movements and impaired consciousness, but unlike epileptic seizures, these typically have prominent proximal limb and truncal movements, side-to-side head movements, crying, and eye closure with resistance to passive opening. Epileptic seizures coexist in at least 10% of patients with nonepileptic seizures [10]. The EEG during the event shows a normal waking background (despite impaired consciousness) which may be obscured by movement artifact [11].

Panic Attacks

Panic (anxiety) attacks are episodes of fear, apprehension, or terror accompanied by hyperventilation, palpitations, chest pain, feeling faint or dizzy, paresthesia (around the mouth and fingers), and a feeling of choking that lasts several minutes. The EEG is typically normal or shows hyperventilation build up. Rarely, insular seizures may resemble panic attacks.

Other Behavioral, Psychiatric, and Psychological Conditions

Many common pediatric behavioral conditions such as day dreaming, inattention, self-gratification (masturbation), eidetic imagery (vivid imaginations), temper tantrums, and out of body experiences mimic seizures. In adults, dissociative states, hallucinations, and occasionally fabricated or fictitious illness may mimic seizures. The EEG is normal during these events [8,12].

Chapter Summary

1. Think of seizure mimics before you diagnose epileptic seizures.
2. Some mimics, such as syncope, transient ischemic attacks (TIA), and migraines, may be associated with EEG abnormalities.
3. Seizure mimics may be neurological, systemic, or psychological.
4. Always confirm if the event in question is consistent with the patient's typical event, as the patient may have more than one type of event.
5. Though uncommon, both epileptic and nonepileptic events may coexist, hence it is important to characterize each of the patient's event types on video EEG.

References

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